

Sepand AliMadadSoltani

Lyon, France | +33 (0)6 49 55 25 42
sepand.a.m.soltani@gmail.com | <https://sepandsoltani.github.io>

Education

Master 2 in Health Engineering: Medical Device Engineering

Polytech Lyon, Claude Bernard University Lyon 1

Lyon, France

2024-2025

- **Concentration:** Medical Image Processing
- **GPA:** 15.31/20
- **Courses:** Magnetic Resonance Imaging (MRI), Segmentation & Registration, Artificial Intelligence in Medical Imaging, Image Reconstruction & Inverse Problems
- **€10,000** Excellence Scholarship awarded for excellent academic background

Bachelor of Science in Electrical Engineering

K.N. Toosi University of Technology

Tehran, Iran

2018-2023

- **Concentration:** Biomedical Engineering
- **GPA:** 16.33/20
- **Courses:** Statistical Pattern Recognition, Signals & Systems, C Programming, Engineering Math, Engineering Probability

Research Experience

Master's Internship: Bayesian Inference of Image-Derived Input Function in Dynamic PET/MR Brain Imaging

CERMEP Laboratory

Lyon, France

March 2025-August 2025

- Developed non-invasive alternative to arterial sampling for PET quantification in 59 patients
- Built an accurate and automatic carotid artery segmentation pipeline on TOF-MR angiography images
- Implemented Bayesian inference model using Markov Chain Monte Carlo (MCMC) for Partial Volume Correction
- Reduced quantification bias significantly compared to standard image-derived methods (MAPE 13% vs 24%)
- Validated method using Monte Carlo PET simulations and cross-site collaboration with Paris-Saclay

Bachelor's Thesis: Interactive and Intelligent Tissue Boundary Segmentation Algorithm

Machine Vision & Medical Image Processing Laboratory (MVMIP), KNTU

Tehran, Iran

January-June 2023

- Built medical image analysis and visualization software in Python
- Implemented intelligent scissors algorithm for semi-automatic boundary detection on 2D image slices
- Integrated the algorithm into the app for fast tissue segmentation with minimal user input
- Incorporated complementary segmentation tools (polygon selection, thresholding, otsu filter)

Skills

- **Programming:** Python, C++, MATLAB, Bash
- **Software and Tools:** GNU/Linux, Git, FMRIB FSL, SPM, 3D Slicer, TPCCLIB
- **Libraries:** Tensorflow, PyTorch, NumPy, pandas, scikit-learn, Matplotlib, ITK, VTK, Boost, Qt
- **Languages:** English (fluent, TOEFL score:101/120), French (Intermediate), Persian (Native)

Work Experience

Sharif University Science & Technology Park

Junior C++ developer

Tehran, Iran

October 2023-July 2024

- Designed and developed a modern interface using the Qt Framework's QML language
- Built and optimized backend logic in C++ to handle large volumes of data efficiently

TECVICO

Medical Image Visualization Software (Freelance Project)

Vancouver, Canada (Remote)

July-September 2023

- Engineered a node-based workflow for composition of custom AI/ML medical image analysis pipeline in Python analysis
- Built a 3D visualization engine using VTK to render volumetric medical data

Conference Posters

Bayesian Image-Derived Input Function Estimation for [¹⁸F]DPA-714 PET: An Evaluation Study

Chloé Teurquety, Sepand Alimadadsoltani, et al.

Submitted Abstract (Under Review)

Projects

Image-based Persian and English Character Sequence Recognition using Recurrent Convolutional Neural Networks(RCNN)

Winter 2023

- Implemented the network based on a paper using the Tensorflow library in Python
- Synthesized images of Persian text of different variety
- Applied data augmentation techniques such as rotating, translating, adding distortion, and adding noise to images
- Successfully trained the model for both languages using the self-made synthesized Persian dataset and public English datasets
- Achieved +85% accuracy for both languages

Deep Learning Analysis of Functional Connectivity Maps in Alzheimer's Disease

Fall 2022-Winter 2023

- Processed raw fMRI/MRI data from the ADNI database using FSL (motion correction, registration) to extract functional connectivity maps
- Evaluated RCNN and CNN models in Tensorflow for feature extraction to classify Alzheimer's disease stages
- Identified critical limitations in using standard deep learning architectures for fMRI analysis

Automated fMRI Pre-processing Pipeline

Summer 2022

- Implemented brain extraction from structural reference MR image
- Implemented fMRI pre-processing including motion correction, slice timing correction, spatial smoothing, and co-registration
- Optimized the pipeline with parallel processing to accelerate computation on the ADNI dataset

References

Dr. Nicolas Costes <i>Research Engineer</i> CERMEP Laboratory, Lyon, France Email: nicolas.costes@cermep.fr Phone: (+33) (0)4 27 85 62 51 Relationship: Master's thesis co-supervisor	Dr. Inés Mérida <i>Research Engineer</i> CERMEP Laboratory, Lyon, France Email: ines.merida@cermep.fr Phone: (+33) (0)4 27 85 62 52 Relationship: Master's thesis co-supervisor	Dr. Kevin Tse-Ve-Koon <i>Master's Program Director & Senior Lecturer</i> University of Claude Bernard Lyon 1 Email: kevin.tse-ve.koon@univ-lyon1.fr Phone: (+33) (0)4 72 43 89 51 Relationship: Academic advisor and teacher
--	--	---